Gate 28
Coordination mechanism study

Native-control equivalence, proposal-order robustness and dependency scheduling

Measured run: 20261103T035936_964985Z | 3 November 2026

Decision. The native MEDVERSE and STATEFUL_FILTER controls produce identical normalized behavior in all 12 existing workflow families. Their tie is structural: the original cycle code distinguishes only LOCAL_FILTER and DENY_ALL. It does not implement a separate stateful-versus-MedVerse mechanism.

A new additive experiment changes only proposal scheduling. Dependency ordering improves completion under delivery-order changes relative to one-pass filtering, but **does not outperform a fair stateful retry scheduler on completion**. Both complete every tested schedule.

Evidence	Measured finding
Deterministic instrumentation / boundary checks	16 / 16 passed
Native-label equivalence audit	12 / 12 families identical
Order experiment	8 known families; 318 effective family-schedule pairs; 5 configurations; 1,590 executions
Stateful retry versus dependency order	318 / 318 completed by each; paired effect sets agree in all 318 pairs
Dependency-ordering efficiency	126 fewer validator calls: 12,060 versus 12,186 (1.03%)
Observed execution-invariant violations	Zero in all five configurations
Artifact preservation	All 903 original files remain byte-identical; existing UI unchanged

What this establishes

Local authorization alone does not ensure that dependency-bound actions complete at the required event. Ordering or fair retry can close that gap. In this finite workflow model, explicit ordering is a modest revalidation optimization, not evidence of uniquely superior institutional harmonization.

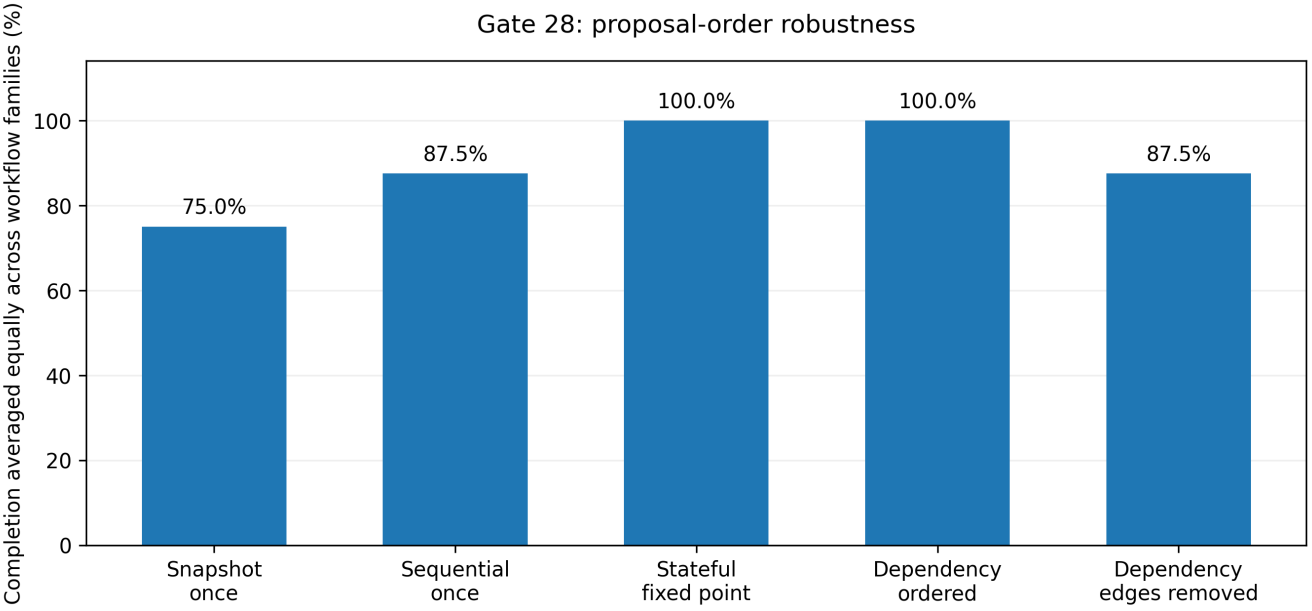
Boundary: developer-authored scenario models, previously seen semantic families and unchanged policy interpretations. No clinical or legal validity, new model generalization, dense-retrieval reproduction or production safety is established. Gate 28 does not replace Gate 26 or 27.

1 | Frozen experimental design

The full 13-item protocol is included as docs/coordination28/PROTOCOL.md. Code, data and schedule specifications were frozen before comparative execution; subsequent report generation does not change them.

Protocol element	Frozen definition
Question	Does explicit dependency ordering add completion or only reduce repeated validation beyond a fair stateful scheduler?
Assumptions	Same five proposers, observations, policy corpus, approvals, execution validator and chronological events. No authority is transferred to the coordinator.
Changed component	Only scheduling: snapshot once; live sequential once; stateful fixed point; dependency order; identical graph code with edges removed.
Data / provenance	Original WF05-WF12 inputs and oracle. Generate 120 role orders x 2 Research action orders, then deduplicate identical actual action sequences before outcomes.
Split / selection	No training or tuning. These eight historical confirmation families are already known. New order perturbations are not independent clinical confirmation.
Leakage audit	Scheduler cannot read expected outcomes, deadlines, family labels or future events. Isomorphic schedule templates are collapsed; semantic overlap is explicitly disclosed.
Oracle	Original positive/negative workflow obligations, plus separate owner, scope, order, duplicate, lineage and audit checks. Not external expert adjudication.
Metrics	Per-family completion, all-schedule robustness, execution invariants, paired effect-set equality, rounds and actual validator-call count.
Pass / fail	Integrity PASS is independent of comparative hypotheses. No threshold or oracle edits after observing results. A strong-control tie is retained.
Raw outputs	Per-schedule and per-family CSVs; compressed proposal-attempt ledger; paired results; normalized native traces; representative full traces; environment and manifest.
Interpretation	Separate policy safety from timely completion and scheduling efficiency. No p-values: schedules are correlated interventions, not independent cases.
Claim boundary	Known finite administrative dependencies, not alternative-workflow synthesis, unknown laws, adversarial LLM reasoning, deployment or clinical safety.
Next dependency	Specify institution-owned alternatives and independent feasible-workflow oracles before testing a broader harmonization contribution.

2 | Completion under changed proposal order



Eight previously seen modeled families; 318 distinct family-schedule pairs per configuration.
Correlated delivery-order perturbations, not 318 independent clinical cases.

Configuration	Macro completion	All-order families	Violations
Snapshot once	75.0%	6/8	0
Sequential once	87.5%	6/8	0
Fair stateful retry	100.0%	8/8	0
Dependency ordered	100.0%	8/8	0
Dependency edges removed	87.5%	6/8	0

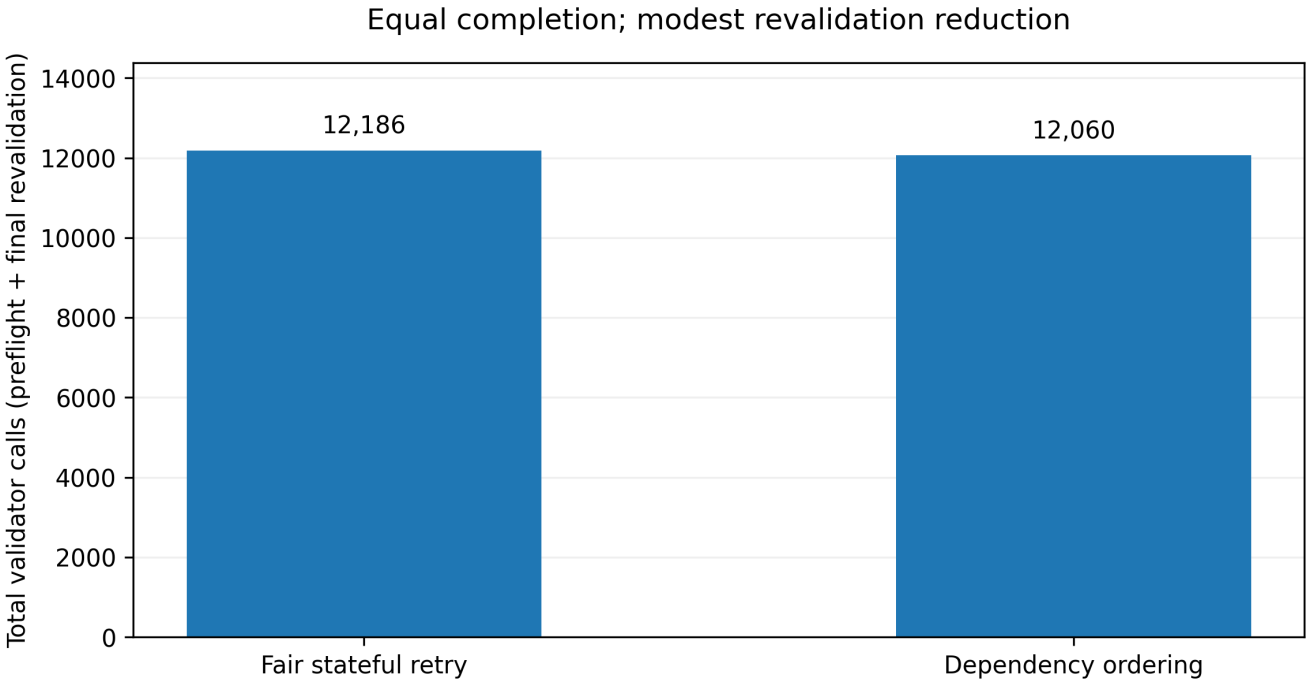
Macro completion is the mean of the eight within-family schedule completion fractions. It does not weight WF05 more heavily simply because it admits more distinct order schedules. The raw pooled rates are retained separately in CSV.

WF05 and WF08 explain the difference: sequential-once completes half of their schedules; dependency ordering and fixed-point retry complete all. The other six families satisfy their modeled obligations under every tested schedule.

Removing only the dependency edges reproduces sequential-once behavior in all 318 paired effect-set comparisons. This isolates the scheduling contribution without changing authorization rules.

Do not re-label these as clinical accuracy. The 1,590 executions are repeated evaluations of eight known workflow families. They are not 1,590 independent patient cases.

3 | The strong comparator tie is retained



318/318 schedules completed by both methods. Dependency ordering saves 126 calls (1.03%).
This is an operation-count result, not a measured end-to-end latency advantage.

Hypothesis	Outcome	Interpretation
H1: graph order improves on single pass	Supported descriptively	Macro completion +12.5 percentage points; no observed safety trade-off.
H2: graph order improves completion over fair retry	NOT SUPPORTED	Both 318/318; paired effect sets identical.
H3: fewer checks at equal completion	Supported, small effect	126 calls saved / 12,186 = 1.03%; not a runtime-speed claim.
H4: remove edges only	Supported descriptively	Returns to single-pass completion and effect sets.

The counter records every preflight and final commit revalidation. A method is not credited for fewer checks if it simply fails to complete legitimate work. The valid efficiency comparison here is dependency ordering versus the fair retry method, which reaches the same outcomes.

Maximum rounds per event: two for fair retry and one for graph ordering on this graph. No p95/p99 speedup is claimed from the one-run wall times. External networking, model inference, real institutional review and graph-construction scaling are outside this experiment.

4 | A concrete trace and the scientific diagnosis

Example: WF05-S002

The event chronology is unchanged. The Research release proposal arrives before its Research attestation within that event. At event 5, the required modeled reviews are present, but the attestation action has not yet committed when the release is first examined.

Method	First release event	Required by event	Result
Snapshot once	6	5	Late
Sequential once	6	5	Late
Fair stateful retry	5	5	On time; second pass
Dependency ordered	5	5	On time; first pass
Dependency edges removed	6	5	Late

The deadline is the original benchmark event cursor, not a newly invented clinical time limit. Clinical review and telemetry handling remain satisfied; no prohibited disclosure occurs. The failure is a coordination liveness/timeliness issue, not a policy-authorization bypass.

Separate the three evidence layers

Gate 26: preserved reference-architecture scores on seven semantic archetypes. The user-reported rerun remains the source for that checkpoint.
Gate 27: 30/30 engineering checks and 8/8 completion for both native MedVerse and the stateful label in the supplied user log.
Gate 28: structural equivalence diagnosis and scheduling-only intervention. This clarifies what the earlier comparison could and could not establish.

What remains scientifically open

The current proposer emits a predefined restricted-release path. It does not yet negotiate different institution-proposed data scopes, select among mutually exclusive authorized workflow alternatives, or solve genuine conflicts between alternative institutional goals. A permutation test cannot establish those capabilities.

The next test should compare a candidate harmonizer with a fair stateful selector and an independent constraint planner given the same alternatives, observations and authority conditions. A centralized planner must be allowed to retain institutional action ownership. Equally valid alternative workflows must not be scored as wrong.

A 13-element draft for that next experiment is included, clearly marked NOT IMPLEMENTED / NOT EXECUTED. No claim of completed alternative-workflow synthesis is made.

5 | Reproduction, environment and remaining work

Run the additive gate

From the MedVerse-Agent project directory:

```
python -B medverse/gates/gate28_coordination_mechanism.py
```

For a code/control-equivalence audit only:

```
python -B medverse/gates/gate28_coordination_mechanism.py --stage audit
```

The default run verifies the supplied freeze, executes the full study, and creates a new timestamped evidence directory. Do not run --stage freeze over the existing manifest. The runner needs Python 3.10+ and the standard library; plotting/report regeneration has optional analysis dependencies. The clean -I -S audit also passed without third-party packages.

Saved-model warning is a separate dependency issue

Your console reports scikit-learn 1.8.0 estimators being loaded under 1.9.0. Official scikit-learn documentation states that cross-version loading is unsupported. Reproduce the saved model environment in a separate environment; preserve the original artifacts and do not suppress the warning or retrain simply to make it disappear. Gate 28 loads no saved classifier, so its coordination result does not depend on resolving that warning. [4]

Evidence locations

Path under project root	Content
data/coordination28/	Frozen code/data manifest and deduplicated schedule bank
outputs/gate28_coordination/	Timestamped measured outputs; LATEST_FULL.json identifies this experiment
docs/coordination28/	Full protocol, original-file hashes, user-log reference and next-study draft
coordination28/	Additive schedulers, tests, runner and optional reporting scripts

Sources and provenance

[1] User-supplied Pasted text(4).txt: local Gate 27 and Gate 26 console output; copied as user_run_reference.txt.
[2] Team_MedVerse_Validated_Sandbox_v1.zip: original implementation and frozen workflow records. All 903 original file hashes checked unchanged.
[3] Gate 28 freeze and run 20261103T035936_964985Z: raw results in this package. Findings are developer-authored modeled experiments, not external validation.
[4] scikit-learn, Model persistence, Security & Maintainability Limitations. Accessed 3 November 2026. Official documentation.

Submission boundary: report dependency-aware scheduling and its limitations accurately. Clinical interpretation, Korean legal applicability, original dense embeddings, and genuinely distinct workflow-choice harmonization remain separate validation requirements.